higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL SURVEYING N5

28 MARCH 2019

This marking guideline consists of 4 pages.

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QUESTION 1

1.1	1.1.1	C
	1.1.2	A
	1.1.3	E
	1.1.4	D
	1.1.5	B

(5 × 1) (5)

1.2	1.2.1	True
	1.2.2	False
	1.2.3	True
	1.2.4	False
	1.2.5	True

(5 × 1) (5)

1.3	1.3.1	B
	1.3.2	D
	1.3.3	A
	1.3.4	A
	1.3.5	D

(5 × 2) (10)
[20]

QUESTION 2

2.1 Used mainly to excavate a uniform bottom of a trench

2.2 Used to measure angles of a sloping side/ground

2.3 Used together with a levelling instrument to measure vertical distances

2.4 Mainly used to mount instruments

2.5 Used to conduct angular measurement either vertical or horizontal in degrees, minutes and seconds

(5 × 2) **[10]**

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QUESTION 3

3.1 Tanø = 0,2/20✓
= $\tan^{-1}$ 0,01✓

Tan0,573 = H/100✓
H =1,0001✓
Slope = 1%✓ (5)

3.2
- Tape must be held horizontally.
- Tape must be held on its correct zero mark.
- The correct/sufficient tension must be applied to the tape.
- Tape must be held on the correct peg.
- View tape vertically over the peg.
- Measure to the centre of a ranging rod. (Any 5 × 1) (5)

3.3
- The gradient of landscapes can be estimated.
- The volumes of excavations or fill can be estimated.
- The ability to see directly between two points can be established by interpolating along a line across contours. (Any 2 × 2) (4)

3.4 Step chaining is conducted on a building site using the following instruments: tapes, plumb bob, pegs and ranging rods.✓ From a peg on the ground the tape is stretched out horizontally✓ – the tape, being comfortably held at waist height. A peg✓ is punched in the ground by using a plumb bob✓ suspended from the horizontal distance being measured. The same operation is repeated until the total distance is measured.✓ (5)

3.5
- Attach the level securely to the tripod.✓
- The tripod feet must be firmly into the ground.✓
- Simultaneously turn adjacent foot screws in opposite directions✓ to adjust the circular bubble to its centre.✓
- Turn the level telescope 90°✓ using the third foot screw to adjust the bubble to its centre.✓ (6)

[25]

QUESTION 4

4.1 Off-sets must be taken from the road reserve✓ to the boundary of the site that is required✓ and a base line parallel to the road must be drawn.✓ These offsets are taken from stipulated chainages on the site plan.✓ The remaining boundaries of the site are measured perpendicularly to the baseline and the required site shape is then set out from the baseline.✓ (5 × 2) (10)

4.2 Three men✓ are required to run in front of the grader, one with a boning rod✓ and the other two holding the fish line on the pegs✓ with the required levels of the particular road layer.✓ The man with the boning rod makes hand signals to the grader operator showing how deep he/she must cut or fill and also whether the required level has been reached.✓ (5)

[15]

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QUESTION 5

5.1

		Join	Sheet	
Station / Peg	**Y**	**X**	**Calculations**	**Direction/ Distance**
A	+10148,250	+10156,324	= $\tan^{-1}$ -148,25✓ -156,324✓✓	D =180°+43°28'53" ✓✓
B	+10000,00	+10000,00	= 43°28'53"✓✓	= 223°28'53"✓✓
	-148,25✓	-156,324✓		
			Check	S = 215,442 m✓✓
			S = 215,442 m × cos223°28'53"	
			= -156,324✓✓	
			OR S = 10,095sin224°32'21"	
			= -148,25	

(15)

5.2

POINT	BACK SIGHT	INTER-MEDIATE SIGHT	FORE-SIGHT	RISE	FALL	REDUCED LEVEL	REMARKS
A	2,300					34,723	TBM 34,723
B	2,545		2,332		0,032✓	34,691✓	
C	2,754		2,233	0,312✓		35,003✓	
D		2,264		0,490✓		35,493✓	
E	2,936		2,342		0,078✓	35,415✓	
F			2,342	0,594✓		36,009✓	TMB 36,002
	10,535		9,249 10,535 -1,286✓	1,396✓	0,11✓ 1,396✓ -1,286	-1,286✓	

(15)
[30]

TOTAL: 100

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